Benchmarking Forecast Rankings for Deployment-Aware Model Selection

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Abstract

Forecasting benchmarks commonly rank models by forecast-side metrics, which is appropriate when the benchmark target is predictive quality. In deployed forecasting systems, forecasts are often consumed through a fixed forecast-to-decision interface: a deterministic evaluation rule that maps forecasts into benchmarked deployed outputs before deployed-side scoring. Under switching or deployment friction, the forecast-metric winner may not be the deployed-side winner even when the interface is fixed. We study this as fixed-interface deployed-selection robustness, a deployment-facing evaluation and benchmark/reporting protocol question for forecasting benchmarks. The paper's main question is Q2: whether forecast-side ranking preserves deployed-side model selection under one fixed interface; Q1 serves only as mechanism support for why divergence can arise. Across a synthetic zero-friction anchor, a forecasting-native event-micro benchmark, a complementary Traffic-Hourly alert family, and inventory corroboration, we find that forecast-side winners can become recurrently deployed-suboptimal as friction increases. These results do not invalidate forecast-side evaluation for prediction; rather, they suggest that forecasting benchmarks aimed at deployment may need to report deployed-selection robustness alongside forecast-side metrics.

1. Introduction

Forecasting benchmarks typically compare models using forecast-side metrics, and those metrics are the right object when the benchmark asks about predictive quality. Some deployment-facing forecasting evaluations, however, consume forecasts through a fixed forecast-to-decision interface such as a thresholding rule or budgeted alert rule. When that interface includes switching or deployment friction, forecast-side ranking can diverge from deployed-side model selection: the model that wins on the forecast metric need not be the model that performs best after the fixed interface is applied.

This paper frames the mismatch as a forecasting benchmark/reporting protocol question. Our main empirical question, Q2, asks whether forecast-side ranking remains reliable for deployed-side model selection under one fixed forecast-to-decision interface. We call the corresponding quantity deployed-selection robustness: the extent to which a benchmark's forecast-side winner also remains the deployed-side winner after the fixed interface and friction are applied. Q1 is only mechanism support: it asks how a frictional interface can separate proposed and realized actions when the proposal path is held fixed. This Q2-first framing keeps forecast-side metrics as prediction measures and asks what deployment-facing evaluation should report alongside them.

Forecasting benchmark work often evaluates forecast quality, calibration, and live predictive performance. Our question is complementary: whether forecast-side ranking remains trustworthy for deployed-side model selection under a fixed forecast-to-decision interface. We do not introduce a forecaster or training objective; the contribution is a benchmark/reporting protocol for deployment-facing evaluation after forecasts are passed through the same fixed interface.

This paper makes three narrow contributions.

- First, we introduce deployed-selection robustness as a forecasting benchmark question: whether forecast-side model ranking remains reliable after forecasts are executed through a fixed interface.
- Second, we document a Q2 divergence pattern: the forecast-side winner becomes recurrently deployed-suboptimal as friction increases under the studied fixed interfaces.
- Third, we provide a Q2-first evidence structure spanning synthetic, forecasting-native event and traffic evidence, and operational corroboration; Q1 serves only as mechanism support.

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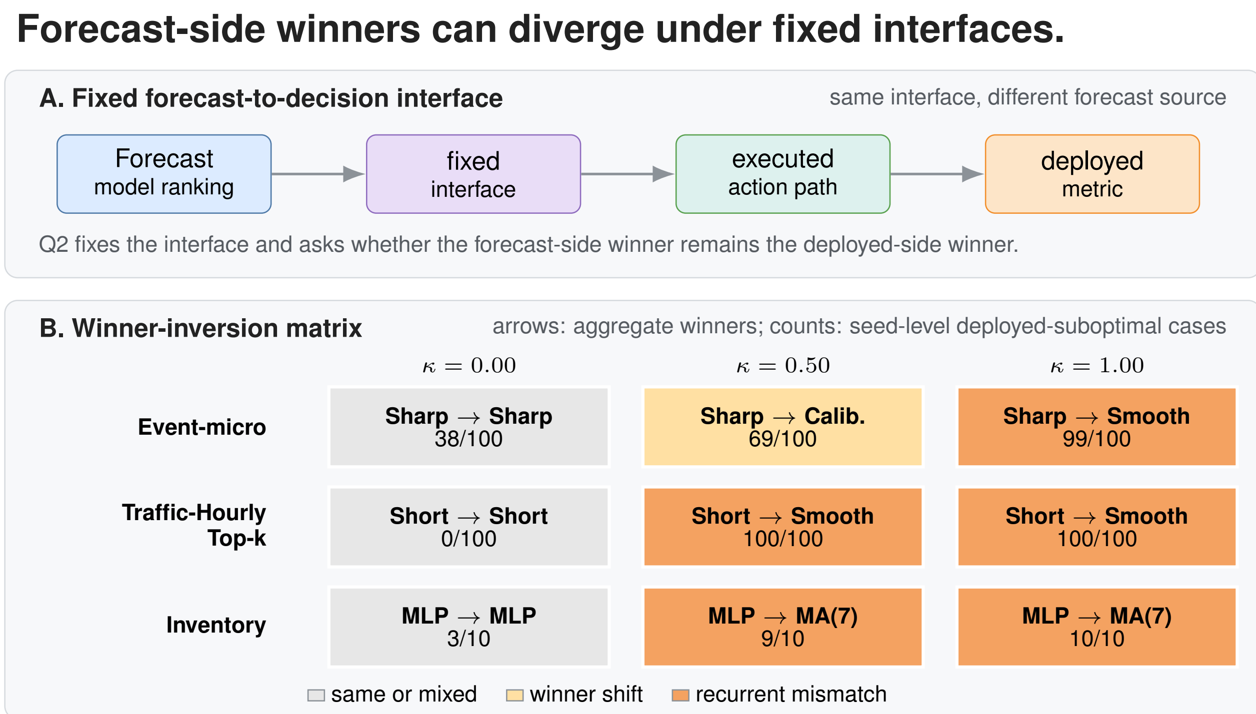

Figure 1. Fixed-interface evaluation can change model selection. Panel A shows the Q2 forecast-to-decision interface, and Panel B reports aggregate forecast-side and deployed-side winners across domains and friction levels. Cell counts denote seed-level deployed-suboptimal cases for the forecast-selected model; darker orange marks recurrent mismatch.

2. Two Evaluation Questions

We distinguish three evaluation objects: forecast-side quality, target-side decision quality, and realized-action quality. Forecast-side quality measures predictive performance, target-side decision quality scores the proposed action path before frictional execution, and realized-action quality scores the executed path after the fixed interface is applied. Q1 holds the proposed action path fixed and varies only the interface, so it is an exact-control mechanism check. Q2 holds the interface fixed and varies the forecaster, so it is the benchmark-selection question. Because Q2 asks whether the forecast-side winner remains the deployed winner, Q2 is the paper's main empirical claim.

3. Evidence Design

We lead with Q2: whether forecast-side winners remain deployed winners under a fixed interface. Event-micro and Traffic-Hourly are the forecasting-native evidence; inventory is operational corroboration.

For inventory Q2, the main text focuses on the 0.50 and 1.00 rows from the fixed five-family comparison; the mixed 0.00/0.25 regime and broader family sweep are reported in the appendix.

4. Results

4.1. Q2: Fixed interface and deployed-selection robustness

Q2 asks the paper's headline benchmark question: under one fixed forecast-to-decision interface, does the forecast-side winner remain the deployed-side winner? Synthetic provides the exact zero-friction sanity anchor. Figure 1 summarizes the main Q2 evidence: event-micro is the primary forecasting-native test, Traffic-Hourly is a second forecasting-native corroboration, and inventory is included only as operational corroboration.

In event-micro, the forecast-side winner remains Reactive sharp across frictions, but the deployed winner shifts to Calibrated baseline at friction 0.50 and Lagged smoother at friction 1.00. Table 1 reports the canonical fixed-threshold interface. The benchmark consequence is recurrent: the forecast-selected model is deployed-suboptimal in 69/100 and 99/100 seeds at these two friction levels. The same qualitative inversion persists under an alternate threshold, under a hysteresis-threshold interface, and when forecast-side ranking is recomputed with log loss.

Friction	Forecast winner	Deployed winner	Agree.	Subopt. seeds
0.00	Reactive sharp	Reactive sharp	0.62	38/100
0.50	Reactive sharp	Calibrated baseline	0.31	**69/100**
1.00	Reactive sharp	Lagged smoother	0.01	**99/100**

Table 1. Event-micro main evidence under the canonical fixed-threshold interface. The forecast-side winner remains Reactive sharp, while moderate-to-high friction changes the deployed winner and makes the forecast-selected model deployed-suboptimal in 69/100 and 99/100 seeds.

A complementary Traffic-Hourly fixed-budget alert family shows the same benchmark consequence under a different forecasting-native task structure. In the selected Top-k setting ($k = 249$), Reactive short is both the forecast-side winner and the deployed winner at zero friction, with exact agreement 1.0. Once switching costs are introduced, however, the forecast-side winner remains Reactive short while the deployed winner shifts to Lagged smoother, and deployed-suboptimality reaches $100/100$ seeds at both friction 0.5 and friction 1.0. A relative-rank Traffic variant shows the same qualitative pattern in the appendix.

Inventory provides operational corroboration rather than the main forecasting-native evidence. At friction 0.50 and 1.00, the forecast-selected Small MLP is deployed-suboptimal in 9/10 and 10/10 seeds, respectively, under the fixed replenishment interface. Table 2 collects the Traffic-Hourly and inventory corroboration rows.

Domain	Friction	Forecast winner	Deployed winner	Subopt. seeds
Traffic-Hourly Top-k	0.00	Reactive short	Reactive short	0/100
Traffic-Hourly Top-k	0.50	Reactive short	Lagged smoother	100/100
Traffic-Hourly Top-k	1.00	Reactive short	Lagged smoother	100/100
Inventory	0.50	Small MLP	Moving average (7)	9/10
Inventory	1.00	Small MLP	Moving average (7)	10/10

Table 2. Cross-domain corroboration under fixed interfaces. Traffic-Hourly provides a second forecasting-native task family, and inventory provides operational corroboration; both show recurrent deployed-side misselection at moderate-to-high friction.

Q1 mechanism support. Q1 explains why divergence can arise; Q2 is the actual benchmark-selection claim. Holding the proposed action path fixed, synthetic and inventory show that frictional interfaces can separate proposed and realized actions. Synthetic yields an increasing proposal-versus-realization gap away from zero friction, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. Q2 then shows when that mechanism produces deployed misselection under one fixed interface.

Appendix robustness. Appendix reports threshold, reranking, and second-interface checks for event-micro, together with the Traffic relative-rank redesign, broader inventory sweep, and load-following check. These appendix checks are consistent with the main qualitative conclusion: under a fixed interface, a forecast-side winner can remain forecast-best while becoming recurrently deployed-suboptimal in moderate-to-high friction regimes.

Why forecast ordering need not be preserved. Proper forecast-side scores evaluate predictive quality, not the composed object obtained after forecasts are mapped to actions and executed through a frictional interface. If a forecaster's forecast-side advantage is smaller than the extra friction-induced loss created by its implied action path, the deployed ordering can reverse even under a fixed interface. This is why Q1 is useful as mechanism support and why Q2 is the actual benchmark-selection question.

5. Discussion, Related Work, and Limitations

Related work. Recent forecasting benchmarks span live AI-forecasting settings, broad time-series archives, and foundation-model evaluations (Karger et al., 2025; Yang et al., 2025; Aksu et al., 2024; Godahewa et al., 2021; Das et al., 2024; Williams et al., 2025). These efforts have made forecasting evaluation itself a central research problem, but they primarily compare models by forecast-side accuracy, calibration, or leaderboard performance. Our question is different: under one fixed forecast-to-decision interface, does the forecast-side winner remain the deployed-side winner? We therefore treat deployed-selection robustness as a benchmark reporting problem rather than as a new forecasting model or optimizer, with variance-aware benchmark reporting providing the closest methodological motivation (Bouthillier et al., 2021).

This differs from proper scoring rules, decision-maker-perspective forecast evaluation, decision-focused learning, and predict-then-optimize, which either evaluate forecast quality/value or modify training/optimization objectives for downstream tasks (Gneiting and Raftery, 2007; Murphy, 1993; Raeth and Ludwig, 2025; Donti et al., 2017; Elmachtoub and Grigas, 2022; Mandi et al., 2024). Here, forecasters are fixed, and only the evaluation object changes.

Discussion and limitations. Our claim is evaluative and deliberately narrow. We do not claim that forecast-side metrics are invalid for prediction, and we do not seek a single reporting rule for all forecasting applications. Event-micro provides the main forecasting-native evidence, Traffic-Hourly provides a second forecasting-native task family, and inventory provides operational corroboration, but this is still not a broad benchmark suite. Broader inventory sweeps are reported in the appendix, and load-following is secondary; the strongest main-text evidence comes from the moderate-to-high friction settings studied here. Our results therefore support a reporting recommendation: deployment-facing forecasting benchmarks should test deployed-selection robustness under fixed interfaces alongside standard forecast-side metrics.

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A. Event-Micro Robustness Package

Because event-micro provides the paper's main forecasting-native Q2 evidence, this appendix records the first domain-specific robustness package for that benchmark. We construct a minimal forecasting-native binary-event setting evaluated over 100 seeds, in which each forecaster emits event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. The main text uses Brier as the canonical forecast-side criterion. This appendix records the full six-row Brier table plus an alternate-threshold check, a log-loss reranking check, and a second-interface hysteresis check.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap
0.00	Reactive sharp	Reactive sharp	0.62	0.003
0.05	Reactive sharp	Reactive sharp	0.60	0.002
0.10	Reactive sharp	Reactive sharp	0.58	0.002
0.25	Reactive sharp	Calibrated baseline	0.51	0.003
0.50	Reactive sharp	Calibrated baseline	0.31	0.011
1.00	Reactive sharp	Lagged smoother	0.01	0.057

Table A1. Forecast-side vs. deployed-side model selection in the full six-row event-micro benchmark summary. Under the canonical fixed-threshold interface, deployed-suboptimal counts rise from 38/100 seeds at zero friction to 99/100 at friction 1.00.

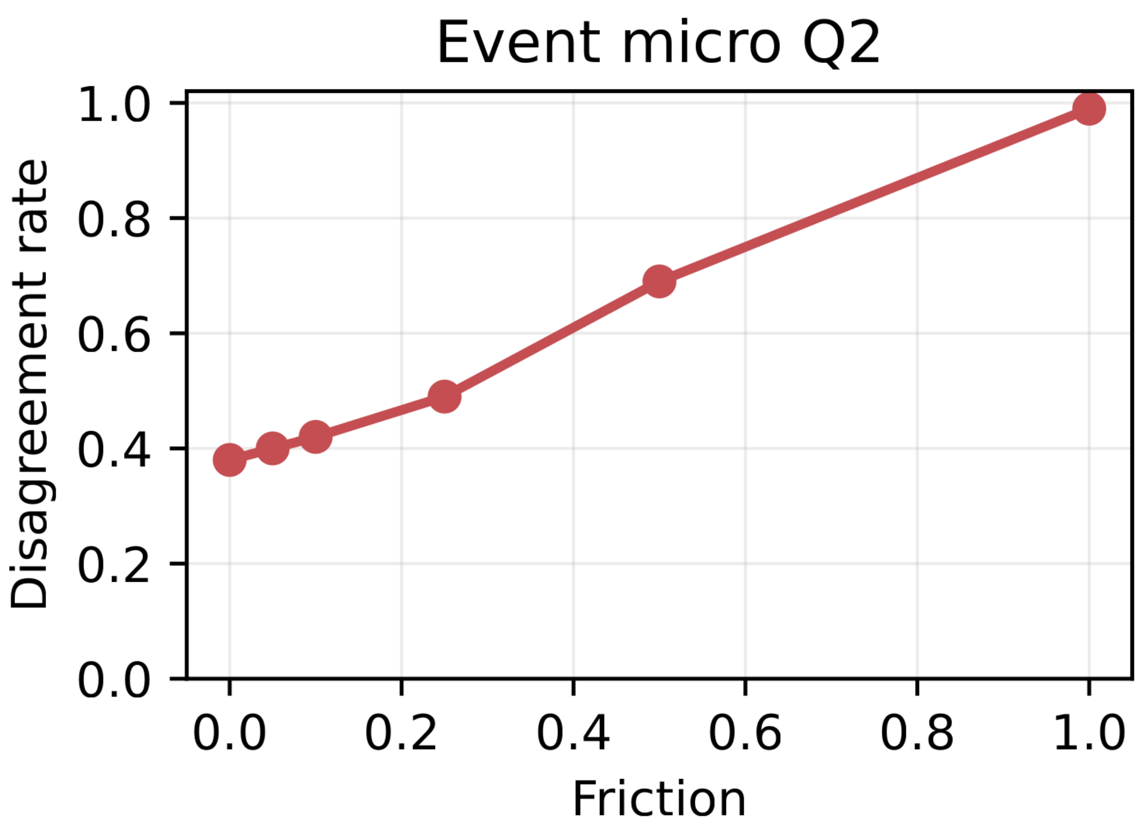

Figure A1. Ranking mismatch in a minimal event-forecasting micro-benchmark. Disagreement between forecast-side and deployed-side model selection is lowest at zero and very low friction and increases as switching frictions rise.

Threshold	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
tau=0.55	0.00	Reactive sharp	Reactive sharp	0.62	0.003	0.000	38/100
tau=0.55	0.50	Reactive sharp	Calibrated baseline	0.31	0.011	0.008	69/100
tau=0.55	1.00	Reactive sharp	Lagged smoother	0.01	0.057	0.053	99/100
tau=0.50	0.00	Reactive sharp	Reactive sharp	0.56	0.002	0.000	44/100
tau=0.50	0.50	Reactive sharp	Calibrated baseline	0.23	0.014	0.011	77/100
tau=0.50	1.00	Reactive sharp	Lagged smoother	0.02	0.061	0.059	98/100

Table A2. Alternate-threshold robustness for the event-micro benchmark. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when the fixed threshold is changed from $\tau = 0.55$ to $\tau = 0.50$.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.002	0.000	38/100
0.50	Reactive sharp	Calibrated baseline	0.26	0.013	0.010	74/100
1.00	Reactive sharp	Lagged smoother	0.01	0.060	0.057	99/100

Table A3. Log-loss robustness for the event-micro benchmark at the canonical threshold. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when forecast-side ranking is recomputed with log loss instead of Brier.

Interface	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
Hysteresis threshold	0.00	Reactive sharp	Reactive sharp	0.55	0.003	0.000	45/100
Hysteresis threshold	0.50	Reactive sharp	Calibrated baseline	0.45	0.005	0.001	55/100
Hysteresis threshold	1.00	Reactive sharp	Lagged smoother	0.18	0.022	0.020	82/100

Table A4. Second-interface robustness for the event-micro benchmark. Under a fixed hysteresis-threshold interface with $\delta = 0.05$, the same qualitative winner drift and deployed-suboptimality pattern remains visible, though the magnitudes are smaller than under the canonical fixed threshold.

Across this appendix robustness package, the same qualitative winner-drift and recurrent deployed-suboptimality pattern persists in a minimal forecasting-native event-micro benchmark evaluated over 100 seeds.

A.1. Interface-Aware Event-Micro Stats Addendum

This addendum records interface-specific recurrence and gap summaries for the hysteresis-threshold check without changing the main cross-domain recurrence package.

Interface	Friction	Deployed-suboptimal seeds / total	Share	95% exact CI	One-sided binomial p
Fixed threshold	0.50	69/100	0.69	[0.605, 1.000]	¡0.001
Fixed threshold	1.00	99/100	0.99	[0.953, 1.000]	¡0.001
Hysteresis threshold	0.50	55/100	0.55	[0.463, 1.000]	0.184
Hysteresis threshold	1.00	82/100	0.82	[0.745, 1.000]	¡0.001

Table A5. Interface-aware recurrence tests for event-micro deployed-suboptimality. The canonical fixed threshold shows stronger recurrence, while hysteresis preserves the same direction with smaller magnitudes.

The aggregate recurrence table reports the seed-level pattern directly, replacing the dense seed-strip visualization.

Interface	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Fixed threshold	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Fixed threshold	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Hysteresis threshold	0.50	0.005	[0.004, 0.006]	0.001	[0.000, 0.003]
Hysteresis threshold	1.00	0.022	[0.019, 0.026]	0.020	[0.015, 0.025]

Table A6. Interface-aware bootstrap intervals for the event-micro deployed gap. Positive intervals remain visible under both interfaces, with smaller gaps under hysteresis.

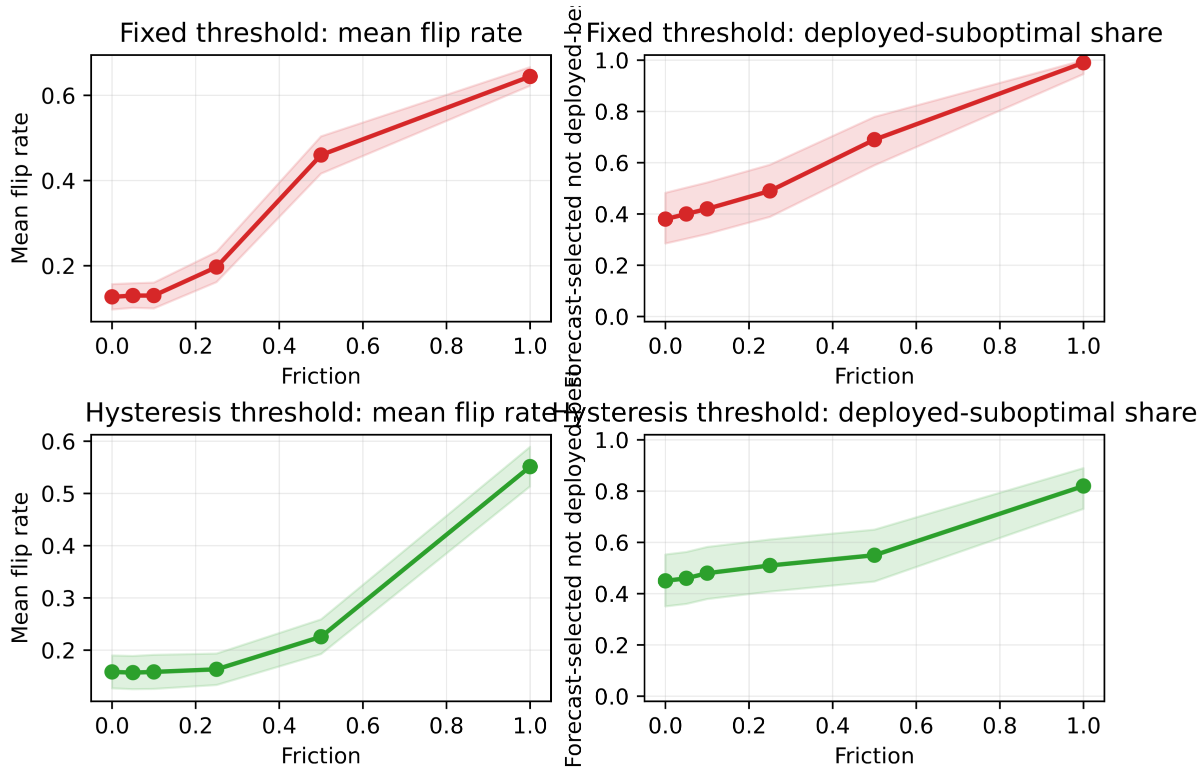

Figure A2. Event-micro robustness across interfaces. Fixed-threshold and hysteresis interfaces show the same direction: disagreement and deployed-suboptimal share rise with friction, while hysteresis attenuates the effect.

B. Traffic-Hourly Forecasting-Native Redesign Families

B.1. Fixed-Budget Top-k Alert

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Lagged smoother	0.00	7.016	6.977	100/100
1.00	Reactive short	Lagged smoother	0.00	24.651	24.473	100/100

Table A7. Traffic-Hourly Top-k corroboration under a fixed interface. Agreement is strongest at zero friction and weakens at higher friction, where the deployed winner shifts away from the forecast-side winner.

In the selected Top-k setting, Reactive short is the forecast-side winner and the deployed winner at zero friction. At frictions 0.5 and 1.0, the deployed winner shifts to Lagged smoother even though Reactive short remains Brier-best and continues to win on the forecast side. The qualitative reading is the same as in the main text: under a fixed interface, the more reactive forecast-optimal family need not remain deployment-optimal once switching is penalized.

B.2. Relative-Rank Traffic Variant

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Calibrated baseline	0.00	10.103	10.122	100/100
1.00	Reactive short	Calibrated baseline	0.00	25.205	25.231	100/100

Table A8. Traffic-Hourly relative-rank corroboration. The same qualitative separation between forecast-side and deployed-side selection appears under fixed-budget set action.

The relative-rank target shows the same qualitative separation under a different forecasting-native label construction. Reactive short remains the forecast-side winner, while Calibrated baseline becomes the deployed winner once switching costs are introduced, so we report this variant as an appendix forecasting-native check. The alternate $m = 0.15$ variant is also strong in the full report but is omitted from the compact table to keep the appendix narrow.

C. Inventory Operational Corroboration

This appendix records the full five-family inventory summary that underlies the compact corroboration rows in the main text. The mixed zero and low-friction rows are reported here, while the main text uses only the moderate-to-high friction rows to avoid overstating the zero-friction story.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Small MLP	0.70	0.022	3/10
0.25	Small MLP	Linear AR	0.70	0.053	3/10
0.50	Small MLP	Moving average (7)	0.10	0.308	9/10
1.00	Small MLP	Moving average (7)	0.00	1.187	10/10

Table A9. Inventory corroboration under a fixed replenishment interface. The zero/low-friction regime is mixed, while moderate-to-high friction shows recurrent deployed-side misselection.

D. Expanded Inventory Sweep

The stronger-baseline sweep is included as a fairness check for the inventory corroboration. It contextualizes the inventory corroboration; the main claim is supported by the fixed-interface selection comparison in the main text.

Domain	Role	Status	Seeds	Zero flip	Zero rho	Best +fric.	Best +flip	Strongest pair	Flip share	Note
Synthetic	required	clears stricter gate	20	0.000	1.000	0.2500	0.500	Linear AR / Reactive heuristic	1.000	passed stricter gate
Inventory	required	mixed zero row	10	0.120	0.840	1.0000	0.500	Small MLP / Moving average (7)	1.000	does not clear stricter gate

Table A10. Expanded same-interface inventory sweep. The table records broader-family status checks for inventory robustness.

Detailed tuning grids and representative selection are summarized compactly; the robustness table reports the resulting fixed-interface comparison.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Gradient-boosted trees	0.60	0.043	4/10
0.25	Small MLP	Gradient-boosted trees	0.60	0.059	4/10
0.50	Small MLP	Gradient-boosted trees	0.50	0.128	5/10
1.00	Small MLP	Moving average (7)	0.10	0.684	9/10

Table A11. Expanded-family inventory Q2 robustness summary for the broader comparison set.

The mixed zero/low-friction rows do not overturn the narrower main claim from the five-family inventory comparison.

E. Load-Following Secondary Check

Load-following is a secondary operational check. It tests whether forecast-side and deployed-side selection can diverge outside the event-micro and Traffic-Hourly settings.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Linear AR	Small MLP	0.70	0.000	0.000	3/10
0.25	Linear AR	Small MLP	0.30	0.001	0.000	7/10
0.50	Linear AR	Small MLP	0.30	0.001	0.001	7/10
1.00	Linear AR	Small MLP	0.20	0.002	0.002	8/10

Table A12. Load-following secondary check. Near-zero friction remains mixed, while positive-friction regimes can favor a different deployed winner.

The result is consistent with the narrower reading used throughout the paper: deployed-side selection can differ from forecast-side selection under friction, but this domain is not used as primary evidence.

F. Q1 Mechanism Support

Q1 serves only as mechanism support for why Q2 divergence can arise. Holding the proposed action path fixed, synthetic and inventory show how a fixed interface can separate proposed and realized actions once friction is introduced.

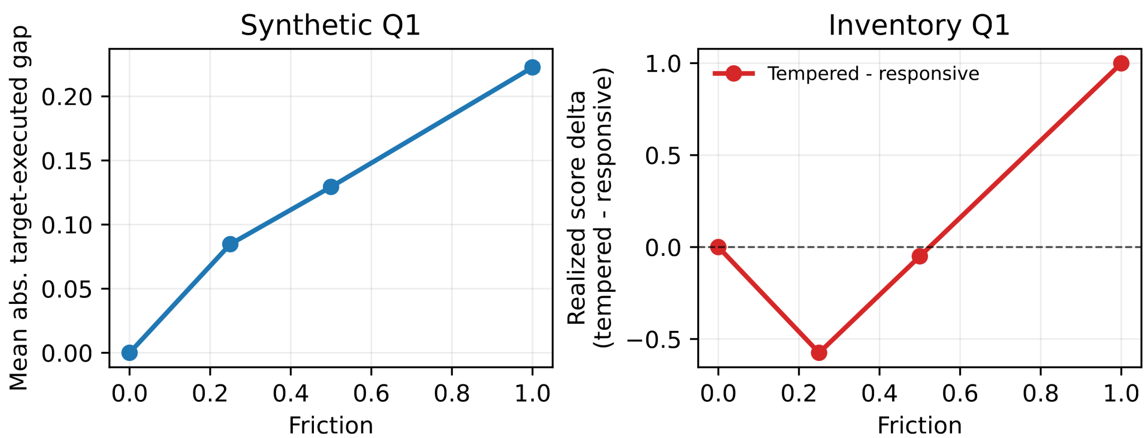

Figure A3. Q1 mechanism support. Even with the same proposed actions, a fixed frictional interface can alter realized actions and realized score; this explains how Q2 ranking divergence can arise.

G. Additional Uncertainty and Seed Recurrence

This appendix collects cross-domain seed-level reliability checks for the main Q2 rows. These appendix additions do not overturn the narrower main-text claim.

Domain	Friction	Deployed-suboptimal seeds / total	Share	95% exact CI	One-sided binomial p
Event micro	0.50	69/100	0.69	[0.605, 1.000]	¡0.001
Event micro	1.00	99/100	0.99	[0.953, 1.000]	¡0.001
Inventory	0.50	9/10	0.90	[0.606, 1.000]	0.011
Inventory	1.00	10/10	1.00	[0.741, 1.000]	¡0.001
Load-following	0.25	7/10	0.70	[0.393, 1.000]	0.172
Load-following	0.50	7/10	0.70	[0.393, 1.000]	0.172
Load-following	1.00	8/10	0.80	[0.493, 1.000]	0.055

Table A13. Seed-level recurrence tests for deployed-suboptimality. Event-micro and inventory show the strongest majority recurrence; load-following remains directional.

Aggregate recurrence tests and uncertainty bands summarize the seed-level pattern without the dense seed-strip visualization.

Domain	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Event micro	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Event micro	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Inventory	0.50	0.308	[0.186, 0.443]	0.227	[0.176, 0.484]
Inventory	1.00	1.187	[0.995, 1.390]	1.162	[0.969, 1.389]
Load-following	0.25	0.001	[0.000, 0.001]	0.000	[0.000, 0.001]
Load-following	0.50	0.001	[0.000, 0.002]	0.001	[0.000, 0.002]
Load-following	1.00	0.002	[0.001, 0.005]	0.002	[0.000, 0.003]

Table A14. Bootstrap intervals for deployed-gap rows. Event-micro and inventory provide the strongest positive-gap support; load-following remains directional.

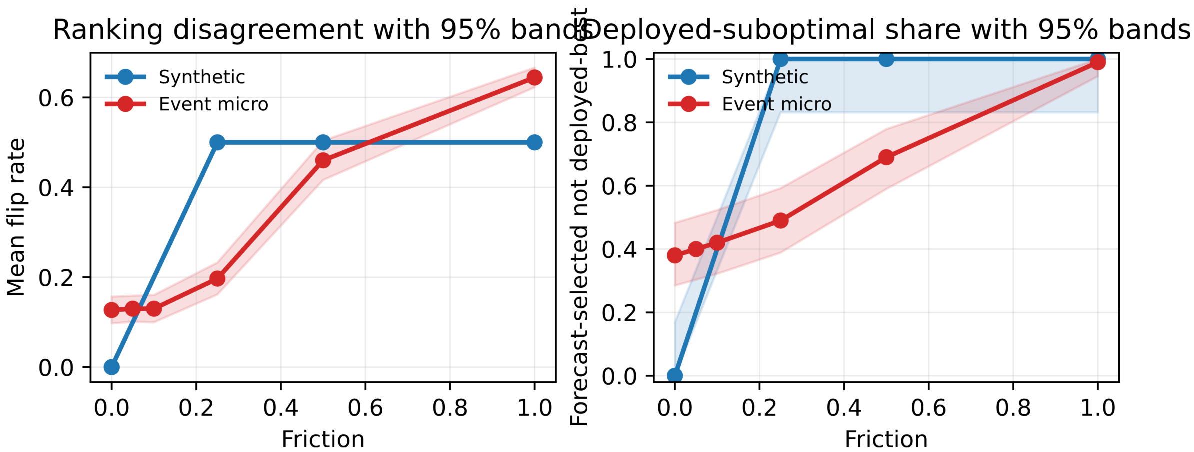

Figure A4. Q2 uncertainty bands for synthetic and event-micro. Panels show mean flip-rate bands and exact 95% binomial intervals for deployed-suboptimal share.